

IBRAHEEM EL SHEIKHA

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🌐 [Ibraheem El Sheikha](#)

📁 [Personal Portfolio](#)

EDUCATION

University of Toronto

Computer Engineering (BASc) & PEY Co-op

Sept. 2023 – Apr. 2027

Toronto, Canada

Relevant courses: *Computer Organization (RISC-V Assembly, Embedded C), Digital Systems (Verilog, FPGA programming), Programming Fundamentals (C++), Computer Fundamentals (C), Operating Systems (Linux), Data Structures and Algorithms, Digital Electronics, Computer Hardware, Embedded Systems, Computer Networks*

Top 30 ECE Student (2025–2026)

Dean's Honours List (2023, 2024, 2025, 2026)

TECHNICAL & LEADERSHIP SKILLS

Software: C, C++, Python, RISC-V Assembly, HTML/CSS, MATLAB, Simulink

Hardware: Verilog, FPGA Programming, HDL Synthesis, ModelSim, Arduino, Circuit design/analysis, Oscilloscope, Multimeter, LTSpice, Altium, KiCAD, Soldering, PyVISA

Leadership Skills: Teamwork, Communication, Collaboration, Problem-solving, Analytical thinking

EXPERIENCE

FPGA Software Engineering Intern | Microchip Technology

May 2026 – Present

- Develop and maintain **C++ CAD/EDA** software for **FPGA design and synthesis** flows, working with **Verilog**, the **Verific elaborator**, **TCL scripts**, and internal tools.
- Contributed to **synthesis optimizations** that reduced **FPGA area utilization** by up to **30%** on targeted designs, improving synthesis quality of results (QoR).

Software Research Intern | University of Toronto ECE

May 2025 – Jun. 2026

- Developed **PiazzaPlus**, a **Chrome extension** for Piazza with a **Flask backend** that enhanced Piazza by enabling students to discover relevant discussions faster through **hybrid semantic + BM25** search.
- Improved** search quality by **120% in MAP** and **152% in MRR** over Piazza's native search.
- First author and presenter** of research paper published in **ASEE 2026** conference proceedings
- Awarded** Edgar McAllister Fellowship (**1 of 2 recipients** from all 1st to 3rd year ECE applicants at UofT)

PROJECTS

STM32 Perimeter Defense System | Embedded C, STM32, ESP32, Python

Mar. 2026 – Apr. 2026

- Designed and built an **automated radar turret** for perimeter mapping and target tracking, integrating **STM32 and ESP32 microcontrollers** with **ultrasonic and infrared sensors**.
- Implemented **embedded control logic** for continuous scanning, tracking, and UART telemetry; achieved **90% accuracy** classifying friendly and enemy infrared-encoded signals.
- Built a **Python/Matplotlib visualizer** that parsed live UART data and rendered a real-time perimeter map.

Embedded C Reversi | Embedded C, RISC-V Assembly

Mar. 2025 – Apr. 2025

- Developed a **real-time** multiplayer Reversi game in **embedded C** on a **RISC-V processor** running on the **DE1-SoC FPGA**, interfacing a PS/2 mouse and VGA display through **memory-mapped ports**
- Implemented and optimized a **60 FPS VGA graphics** pipeline via direct **memory-mapped framebuffer** and register access, rendering dynamic game elements: scoreboard, current-player indicator, legal-move highlights, and winner display
- Developed comprehensive **firmware game logic** (move validation, score tracking, win-condition detection), demonstrating seamless **hardware–software co-design** and efficient resource utilization

Custom RISC-V CPU | Verilog, ModelSim, Quartus, FPGA Programming

Oct. 2024 – Dec. 2024

- Implemented and debugged a **RISC-V-inspired CPU from scratch** using **Verilog** and **Quartus Prime Lite**, debugged in **ModelSim**
- Designed and implemented a robust **VGA display interface**, including **memory arbitration logic** and direct framebuffer management
- Handles **core instructions** (addi, add, sub, xor, and, or, sw, lw, beq, bne)